

Technology Stack

Date	6 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification/Purpose
1	Frontend/Client-Side	HTML, Jinja2, Bootstrap 5.3.2, Chart.js, JavaScript	Used to build a responsive and interactive user interface. Jinja2 dynamically renders application data, Bootstrap provides responsive styling, JavaScript handles client-side interactions, and Chart.js displays dashboard visualizations.
2	Backend/Server-Side	Python 3, Flask, Flask-CORS	Used to handle application routing, form submissions, prediction requests, API endpoints, and communication between the machine learning model and database.
3	Database/Data Storage	SQLite, SQLAlchemy ORM	Used to store and manage user data, prediction records, and report information. SQLAlchemy provides efficient interaction with the SQLite database through ORM-based models.
4	Cloud/Hosting/Deployment	Render, Gunicorn	Render is used to host and deploy the application, while Gunicorn serves the Flask application efficiently in the production environment.
5	Version Control & CI/CD	Git, GitHub	Used for source code version control, repository management, change tracking, and collaboration among team members. No automated CI/CD pipeline is implemented.
6	Third Party APIs/Other Tools	scikit-learn, XGBoost, pandas, NumPy, joblib, FPDF	Used for machine learning-based crop prediction, data preprocessing, model training and persistence, and PDF report generation. No external third-party APIs are used.

